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Firebreak and Fuelbreak



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Synonyms

Barriers; Firebreak; Fuelbreak; Fuel managed areas

Definition

Firebreak and fuelbreak are fuel-managed areas meant to stop or hamper fire propagation, respectively. Firebreaks are parcels of land, linear in shape, where fuels are totally removed down to mineral soil through a combination of physical treatments (thinning, mechanical clearing, prescribed burning, slashing, mastication, mowing, plowing). Fuelbreaks are parcels of land, linear or in blocks, covered by vegetation where fuels are reduced in both volume and flammability by physical treatments and land management practices such as grazing, game management, or silvicultural treatments favoring less-flammable vegetation, so that fires burning into them can be controlled more effectively and safely. In the event of fire, both firebreaks and fuelbreaks may serve as a control line from which to carry out suppression operations.

Introduction

The mitigation of negative effects of wildfires on ecosystem services and human assets can be achieved by reducing the fire rate of spread and the intensity of the fire front. As wildfires spread through a landscape where vegetation is available for combustion, a key strategy to reduce fire intensity, fire perimeter growth, and mitigate the fire severity is to limit the biomass available for combustion by the management of vegetation fuels in specific areas (Agee et al. 2000). Firebreaks and fuelbreaks are parcels of land, linear or in blocks, where the flammability of fuels is reduced by total or partial removal through physical treatments (thinning, mechanical clearing, prescribed burning, slashing, mowing, etc.) or land management practices (grazing, game management, use of alternative crops or favoring less-flammable

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vegetation by artificial or natural regeneration). Firebreaks and fuelbreaks can be distinguished in (i) linear firebreaks and fuelbreaks, (ii) area-wide fuelbreaks treatments, and (iii) fuelbreaks at the wildland-urban interface.

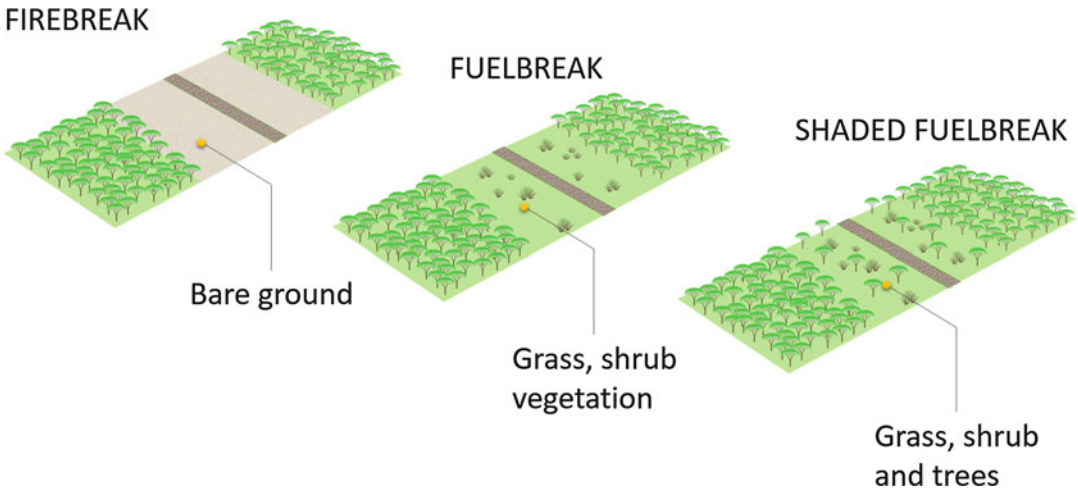
Linear Firebreaks and Fuelbreaks

Linear firebreaks, fuelbreaks, and shaded fuelbreaks are terms referring to fuel treated areas with different characteristics (Fig. 1). Linear firebreaks are strips of land cleared to the mineral soil, while fuelbreaks maintain the herbaceous and shrub vegetation, and a shaded fuelbreak maintains the tree cover. Linear fuelbreaks and shaded fuelbreaks are more suited in mountainous territories (i.e., steep slopes and rough morphology) where the maintenance of the vegetation cover is advantageous to limit soil erosion.

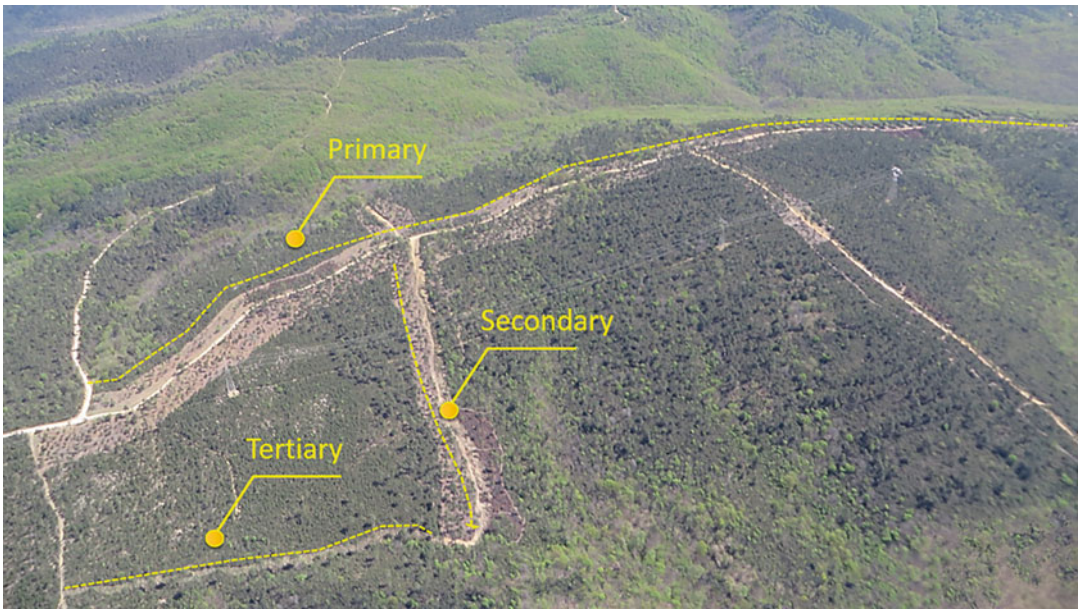
Linear firebreaks and fuelbreaks are classified according to their decreasing width in primary, secondary, and tertiary (Fig. 2) with different objectives, technical specifications, and management approaches for each type. Recommendations regarding width are variable across geographical areas and related fire management system. For the primary ones, minimum width recommendations range from 30 to 100 m. Both secondary and tertiary networks are recommended to range from 3 m to more than 50 m wide, depending on the geographical areas and specific situations. In territories where the wildland-urban interface is widespread, the implementation of large and long firebreaks and fuelbreaks is less suited, as well as in areas where the forest property is mainly private and cadastral land parcels are small. The width can be defined on local expertise (Agee et al. 2000; Bahro et al. 2007), by Forest Fire Management Plans or by a fuel modeling system and a fire behavior prediction software (see the following section of modeling tools). The width of a given fuel discontinuity should at least prevent any material located at the other side of the strip from being ignited due to the heat received through radiation (Agee et al. 2000). As it relates to shape, some of the primary fuelbreaks are aimed at being established as a network creating landscape wide treatments of 500 to 10,000 ha, depending on the region. The others are established as strips or as buffers surrounding, e.g., valuable land or assets.

The primary and secondary network aims at facilitating fire suppression (i.e., making it safer and more effective), promoting the passive protection of assets and fastening the suppression of starting fires, whereas the tertiary network only aims at this latter objective. Other objectives include (i) back-burning (burning out field ahead under safe conditions) to stop wildfires and decreasing fire rate of spread; (ii) modifying fire severity, minimize erosion, and water quality/flow problems; and (iii) protecting valuable assets.

Guidelines for the location of firebreaks and fuelbreaks are based on a wide range of criteria. Overall, the expected fire environments, i.e., fuels (land cover), terrain (topography), and fire weather, are taken into account for defining locations. Rivers, croplands, and grasslands and previously burned areas can be suitable to connect both firebreaks and fuelbreaks. In very steep terrain, ridgelines or valley bottoms are preferred locations for implementation. Other criteria for selecting location include maintenance costs, accessibility (along roads), and fire fighter safety. Knowledge of predominant winds and history of past large fires should also be taken into account (Fig. 3). Indeed, the spatial design of low combustibility linear firebreaks and fuelbreaks is based on anticipating the type of large fires that may affect a specific landscape, by identifying the location of the strategic management points in which fuel management actions would limit fire propagation (Agee et al. 2000). Based on the study of historical large fires in the area and the analysis of the synoptic weather conditions associated with each of those fires, the most frequent fire situations are grouped in typology of fires, i.e., of fire propagation patterns (Duane et al. 2015). Fire behavior modeling can support firebreak and fuelbreak location (see the following section of modeling tools). Standards for design require continuous fuel management between two anchor points of low flammability



Firebreak and Fuelbreak, Fig. 1 Linear firebreaks and fuelbreaks can be distinguished in firebreaks, strips where all vegetation fuels are removed up to the bare soil; fuelbreaks, strips where grass and shrub fuels are maintained; shaded fuelbreaks, strips where grass, shrub, and trees are maintained although with a reduced density

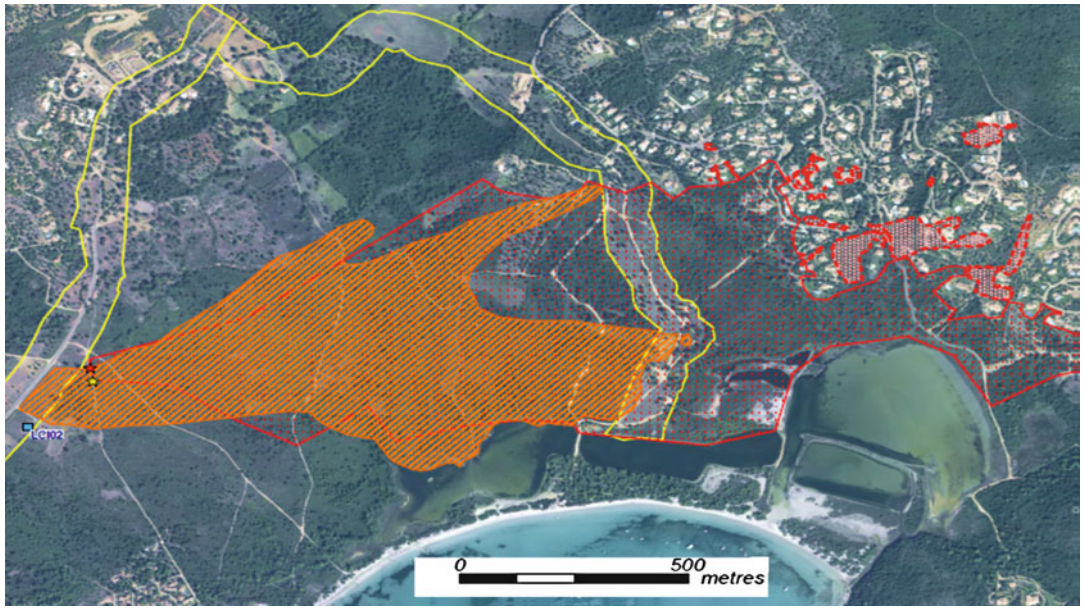


Firebreak and Fuelbreak, Fig. 2 Network of primary, secondary, and tertiary shaded fuelbreaks in Mediterranean pine forests in Southern Europe © Tonarelli L., D.R.E.am Italia

(agricultural areas, less-flammable vegetation), avoiding weak points along the break.

Often linear firebreaks and fuelbreaks are placed along roads, railways, and power lines to control fuel build up on both sides of the network on a maximum width of 50 m. Adjacency to valuable assets (e.g., watersheds, natural parks, archaeological sites) is also a key criterion for defining their location. Forest ownership influences also location as firebreaks and fuelbreaks are implemented mostly in public land.

Fuelbreak management (implementation and maintenance) can be achieved by prescribed



Firebreak and Fuelbreak, Fig. 3 Fuelbreaks (yellow line) located along the major path of recurrent large fires in 2003 (red polygon) and in 2007 (orange polygon) in Corsica, Southern Europe. The fuelbreak was designed after the fire in 2002 and contributed to control the fire in 2007 © Massaiu A., Office National des Foret, France

burning (Fernandes et al. 2013), grazing, mechanical (e.g., mastication) and manual treatments to remove surface shrubs, and thinning to avoid continuity between tree canopies or by maintaining land uses suitable for the objective (e.g., croplands, pastures). Chemical treatments are rarely applied and are favored only along power lines where prescribed burning is limited due to smoke impacts on conductors and fire fighter safety.

Usually, surface fuel load is recommended to be reduced to 2–4 ton/ha. Tree cover specifications range from a minimum of 10% to a maximum of 50% cover. Specifications related to crown base height are set to over 2.5–5 m depending on surface fuels. The need for maintenance depends on site productivity and can be annual up to every 5–6 years.

Area-Wide Fuelbreaks

Area-wide fuelbreaks consist in fuel management taking place over large land areas. They can be implemented using (a) preventive silviculture in forested areas, (b) prescribed burning or grazing, (c) implementing or maintaining annual and permanent crops. The specifications regarding fuelbreaks size are quite variable (1 to 5000 ha), according to the geographical region. Area-wide treatment objectives are decreasing fire growth and intensity, spotting probability and reducing ignitions, and increasing forest resistance and resilience to fire (Fig. 4).

Preventive silviculture is usually integrated within broadest forest management objectives such as increasing soil protection and the mechanical stability of forests or changing even aged forest stands, such as flammable pine plantations, to multispecific uneven aged forests by favoring natural successions and biodiversity (Moreira et al. 2011). Recommended locations include areas with higher fire risk derived from fuel load maps, history of past fires (mainly fire paths and predominant wind direction), topography, climatic regions, proximity to ignition sources, and simulation models of fire spread (see the following section of modeling tools). Usually, area-wide treatments for stand self-protection are restricted to forest areas of high patrimonial



Firebreak and Fuelbreak, Fig. 4 Area-wide fuelbreak (central area of the photo) after a large wildfire in Southern Europe. The reduction of surface fuels and thinning has reduced the fire intensity and severity, compared to the adjacent areas. Indeed, as many trees survived to the fire, the forest has increased the resilience to disturbance © Massaiu A., Office National des Foret, France

or economical values. Less common criteria for defining location include topography, the ability to implement the treatment, grazing areas, and silvicultural objectives set by forest management plans.

As for the type of management, prescribed burning is a common fuel treatment practice in many geographical areas, although in the proximity of urban areas its implementation has higher costs, an increased risk, and greater concern for smoke. This is carried mostly through ground and, in more remote areas, by aerial burning. In some cases, the burning of silvicultural residues following harvesting is implemented. Manual and mechanical treatments are also common, particularly in areas that have good accessibility. Treatments include thinning, pruning, slash removal, conversion from coppice to high forest stands, and implementation of strips of vegetation that have low combustibility, e.g., hardwood or deciduous species. Often techniques are integrated (Fig. 5) and mechanical treatments followed by prescribed burning are considered the most effective in reducing forest flammability (McIver et al. 2012). Grazing management in fuelbreaks is mostly restricted to areas adjacent to linear fuelbreaks. The use of biomass for bioenergy is increasingly used as an additional objective for fuel removal.

Fuelbreaks at the Wildland-Urban Interface

Fuelbreaks at the wildland-urban interface consist in changes to fuel characteristics in parcels adjacent to homes and communities so as to reduce wildfire risk and create safe zones for firefighting (Rigolot et al. 2004; Radeloff et al. 2005).

The recommended fuelbreaks width where fuel buildup is controlled is different in the case of villages (>100 m) versus isolated buildings (>50 m) (Table 1). Distance between tree crowns and stems should be up to 3–4 m and > 6 m, respectively, while the distance from buildings up to 10 m. When selecting tree species around buildings, low flammability species should be



Firebreak and Fuelbreak, Fig. 5 The combination of thinning, fuel mastication, and prescribed burning is considered the most effective fuel management system to reduce hazardous fuels in fuelbreaks

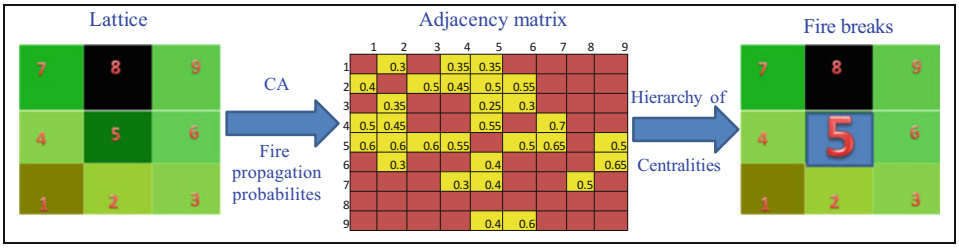
Firebreak and Fuelbreak, Table 1 Technical specifications (e.g., width, tree cover, crown base height) for fuel management at different types of wildland-urban interface (first column). Data has been collected from Gill and Stephens (2009), Miller et al. (2016), Rigolot et al. (2004), Tasmanian Fire Service (2005), Winter et al. (2002), Xanthopoulos et al. (2011)

	Design elements							
Type	Width (m)	Tree cover (%)	Crown base height (m)	Distance between tree crowns (m)	Crowns distance to buildings	Low flammability species	Buffer with no fuel	Surface fuel load (ton/ha)
Villages	> 100	< 35	> 4 or 50% tree height	> 4–6		Yes		< 2 or < 15% shrub cover
Scattered buildings	20–90	< 35	> 2–5 or 50% tree height	> 2.5–4	> 3–5	Yes	1–20 m	< 2 or < 15% shrub cover
Railway	> 2–10	Reduce						Reduce
Roads	> 1–30	Reduce	> 3				2–5	Reduce
Power lines	2–20	Reduce						Reduce
Recreational areas	10–100		> 4 or 50% tree height	> 3–4	> 5		1–3	< 2 or < 50% shrub cover
Industrial areas			> 4 or 50% tree height	> 3–4	> 5		1–2	< 2 or < 50% shrub cover

selected and maintained by tree pruning up to 50% tree height. Surface fuel cover should be less than 20% and 2 ton/ha and less than 50 cm height and continuity disrupted by strips of 2.5 m, distance between shrubs >3 m and no vertical continuity with tree layer, and a tree cover <35%. The access road to the building should be cleared on 10 m on both sides.

Modeling Tools to Assist the Design of Firebreaks and Fuelbreaks

Fire propagation simulation software implement "firebreak" and "fuelbreaks" to test the effect of fuel reduction areas on the fire growth and intensity. These tools enable fire behavior (rate of spread, flame length and fireline intensity), radiation and convection functioning, and spot fire occurrence estimation for specific meteorological, fuel, and topographic conditions (e.g., Finney 2006; Sullivan 2009; Vacchiano and Ascoli 2015). The data obtained from these scenarios (average and extremes) are used to estimate the firebreak width needed to stop the spread of fire by radiation. In fuelbreaks and shaded fuelbreaks, simulation tools are used to estimate the minimum amount of surface fuels allowed to reduce the fire intensity to a desired level for effective fire suppression by fire brigades.



Firebreak and Fuelbreak, Fig. 6 Scheme of the network-based approach for the placement of firebreaks (Russo et al. 2016). Based on the terrain and fuel characteristics, a graph is constructed whose edges are the fire propagation probabilities as computed using a cellular automata model (Alexandridis et al. 2008, 2010). Then centrality measures are computed from the adjacency matrix to find the nodes of the graph (patches of land) which contribute more to the spread of the fire: with black color is depicted an empty cell, i.e., containing no fuel, while intensity of green reflects the density of vegetation. For the particular paradigm, the most "dangerous" node for the spread of the fire is the node 5

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Commonly used simulators such as Farsite (Finney 1998) and FlamMap (Finney 2006) give the option to interactively create "barriers" able to interrupt surface fire propagation. A user can define (using the mouse or loading a map) a segment or a broken line as a "barrier" and also define its depth. It is important to note that in the tools, the "barriers" will block surface fire, but spot fires may be transported over the barrier.

Practical evidence shows that software tools can be successfully used in testing fuel treatments (Stephens 1998), thus acting as decision support systems, i.e., when evaluating the most effective firebreaks (i.e., barrier) for a particular location (Mansourian et al. 2005). For example, a study by Duguy et al. (2007) in the Spanish Mediterranean using Farsite suggested that firebreaks could contribute to reducing the fire size. Fire behavior modeling can support firebreaks and fuelbreaks location. A fire modeling-based approach allows to generate scenarios of fire growth (major fire paths the fire front probably will follow) in a given vegetation, meteorological, and topographical scenario. The data obtained from these simulations can be used to locate the fuel discontinuities that need to be introduced in the landscape in order to minimize the propagation of potential fires. However, in many simulators, the position, the length, and the depth of the firebreaks are defined by the user; thus they cannot be automatically optimized. This represents a serious limitation for the deployment of these systems in designing and managing wildfires. Indeed, while the simulation approach is very useful to decide if a particular choice of the distribution of firebreaks and fuelbreaks is effective or not, alone it does not give any a priori information about what is the best distribution which reduces the hazard risk of forest fire.

Recent studies focused on the mathematical assessment of spatial distribution of firebreaks and fuelbreaks in a landscape (e.g., selective harvesting or prescribed burning) for the management of wildland fires (Palma et al. 2007). In particular, a number of optimization models have been proposed for spatial allocation of fuel treatments (Hof et al. 2000; Wei et al. 2008; Rytwinski and Crowe 2010; Wei 2012). Optimization modeling strategies, based on operational research approach (Minas et al. 2012; Minas et al. 2013), typically require thousands of objective function calculations; thus the solution of the optimization problem is in general highly computationally demanding (Hillier and Lieberman 1990). Moreover, the introduction of oversimplifications to reduce the computational cost poses strong doubts on realistic utility (Rytwinski and Crowe 2010). On the other hand, simulation-optimization approaches (Minas and Hearne 2016; Martell 2007) have been applied for finding the best allocation of firebreaks and fuelbreaks with, for example, the objective of minimizing fire-risk subject to a limit on the total area to be used as fuelbreaks. Although the results presented in Rytwinski and Crowe (2010) suggest

that simulation-optimization is a robust approach to plan fuel treatment mosaics that can reduce fire-risk in a forest, there are still several issues to reduce the computational high cost.

Very recently, based on complex network theory, it has been proposed a computational methodology that addresses the spatial distribution of firebreaks for the inhibition of the spread and size of wildland fires on heterogeneous landscapes (Evaggelidis et al. 2015; Russo et al. 2015, 2016). This is a two-tier approach where the dynamics of fire spread are modeled as a random Markov field process on a directed network whose edge weights are provided by a state-of-the-art cellular automata model (Alexandridis et al. 2008, 2011) that integrates detailed GIS, landscape, and meteorological data. Within this framework, the spatial distribution of firebreaks is reduced to the problem of finding the network nodes among which the fire spreads faster; thus their removal favors the inhibition of the fire propagation (see Fig. 6).

Cross-References

- Fuel Treatments
- Mastication
- Modeling Approaches
- Prescribed Fires
- WUI Risk Assessment at the Landscape Level

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